

Department of Leather Engineering
Khulna University of Engineering & Technology
 B. Sc. Engineering 3rd Year Backlog Examination-2021
LE 3101
Leather Manufacturing Technology IV

Time: 3.0 Hours

Full Marks: 210

N.B. i) Answer any **THREE** questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION-A

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|------|---|----|
| 1(a) | Define leather drying. Briefly state the constant drying rate period during leather drying. | 08 |
| 1(b) | State the terms: (i) Vacuum drying, (ii) Natural drying, (iii) Tunnel drying and (iv) Toggle drying | 08 |
| 1(c) | Describe the following terms in context of leather drying: (i) Equilibrium moisture content, (ii) Bound water and (iii) Free water. | 09 |
| 1(d) | Mention some common defects that may occur during leather drying. Also state their possible remedies. | 10 |
| 2(a) | State the key features of the following leathers: (i) Glove leather, (ii) Football leather and (iii) Suede leather. | 10 |
| 2(b) | State the controlling techniques that should follow for the design of deliming & bating operation for glove leather production. Also formulate a recipe of the deliming & bating for glove leather manufacturing from limed goat skin pelt. | 10 |
| 2(c) | Formulate a recipe from wet-blue to crust for making shoe upper leather of 1.0 mm thickness showing every detail. | 11 |
| 2(d) | How does deacidification differ from shoe upper to garment leather? | 04 |
| 3(a) | Write down the selection criteria of raw materials for the following leathers: (i) Glace-kid leather, (ii) Nappa leather and (iii) Garment leather. | 12 |
| 3(b) | Write on the controlling of liming operation for ideal Glace-kid leather production? Give the proper recipe of liming of Glace-kid leather. | 08 |
| 3(c) | State the specialty of gloss for an deal Glace-kid leather. | 07 |
| 3(d) | What is fur leather? State the principle of the manufacturing of fur leather. | 08 |
| 4(a) | Define upholstery leather. Write down the characteristics of upholstery leather. | 07 |
| 4(b) | Mention the selection criteria for choosing raw materials for upholstery leather. | 06 |
| 4(c) | Formulate a recipe for making upholstery leather from soaking to fat liquoring with every detail. | 15 |
| 4(d) | Define technical leather. Write down the properties of technical leather. | 07 |

SECTION-B

5(a)	Define cationic finish. Cationic finishing is considered as less finishing loading. Explain.	09
5(b)	Illustrate the schematic diagram of the different finishing coats.	08
5(c)	Write down the factors influencing the leather finishing.	09
5(d)	State the following terms: (i) Buffing, (ii) Glazing and (iii) Plating.	09
6(a)	What is conditioning? Why conditioning is done before finishing.	07
6(b)	Write down the working principles of auto spray machine and list out the major controlling parameters.	10
6(c)	Write notes on the following operations: (i) Polishing, (ii) Staking, (iii) Milling and (iv) Snuffing.	12
6(d)	Define buffing. Point out the impact of improper buffing on leather.	06
7(a)	Justify the application of impregnation and adhesion coat with suitable recipe.	11
7(b)	Write the objectives of ground coat with mentioning cautions in choosing chemicals.	08
7(c)	Justify the application of dye staining coat.	08
7(d)	State the criteria of finishing for top coating of leather.	08
8(a)	Write notes on the following finishing chemicals: (i) Plasticizer, (ii) Filler, (iii) Resin binder and (iv) Waxes.	16
8(b)	Write down the advantages and disadvantages of NC lacquer usages in leather finishing.	07
8(c)	Differentiate between acrylic and PU binder.	06
8(d)	Write about the order of finishing season preparation sequentially.	06